THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

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LAB:	**07**
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COURSE CODE:	**CS-2203**
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LAB 07: NETWORK TESTING USING CLI COMMANDS

1. OBJECTIVE

- To gain proficiency in executing foundational Command-Line Interface (CLI) networking utilities.
- To verify host TCP/IP configuration parameters using ipconfig.
- To test end-to-end network reachability and packet round-trip latency using ping.
- To analyze Layer 2 and Layer 3 packet forwarding paths and hop counts using tracert.

2. NETWORK TOPOLOGY & ENVIRONMENT SETUP

The command-line diagnostic tests were executed on a Local Area Network (LAN) simulated in Cisco Packet Tracer. The setup consists of end devices connected through a central Cisco 2960 Switch.

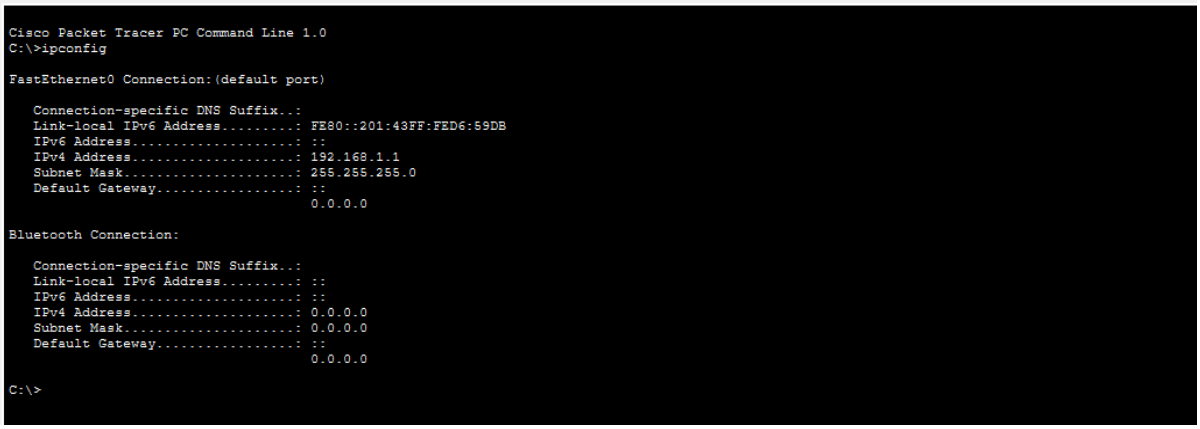
2.1 TOPOLOGY & ADDRESSING SCHEME

- **NETWORK DEVICE:** Cisco 2960 Switch (Switch0)
- **END DEVICES:** Host PCs (PC0, PC1, PC2, PC3) connected via Copper Straight-Through cables.
- **IP Addressing (Subnet 192.168.1.0/24):**
  - **PC0 (HOST TERMINAL): IP** 192.168.1.1 | Subnet Mask 255.255.255.0
  - **PC1 (TARGET HOST):** IP 192.168.1.2 | Subnet Mask 255.255.255.0
  - **PC2 / PC3 (ADDITIONAL NODES):** IP 192.168.1.3 | Subnet Mask 255.255.255.0
  - **DEFAULT GATEWAY:** Left blank (0.0.0.0) as communication is strictly confined to the local switched subnet.

3. COMMAND ANALYSIS & EXECUTION

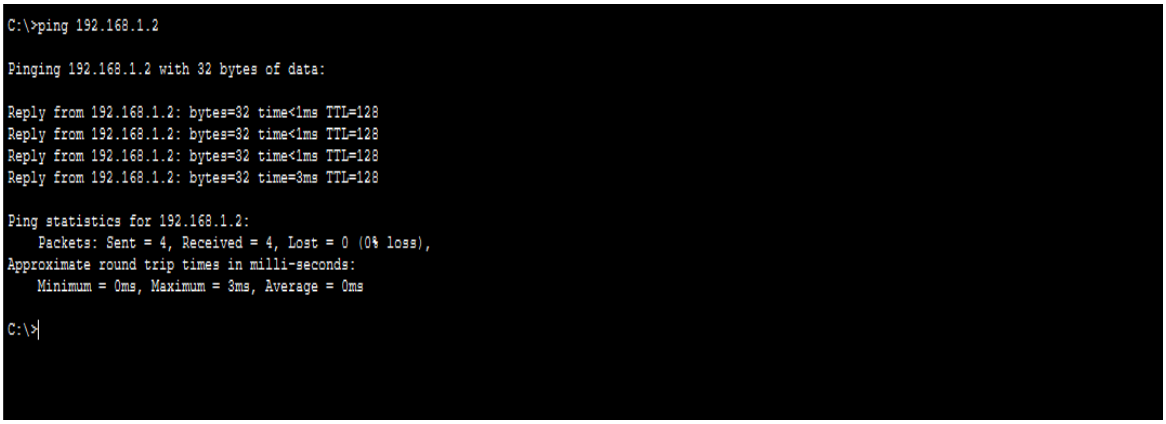
COMMAND 1: IPCONFIG

- **Purpose:** Displays all current TCP/IP network configuration values, including the IP address, subnet mask, and default gateway assigned to active network adapters.
- **Execution:** C:\> ipconfig



COMMAND 2: PING

- **Purpose:** Verifies Layer 3 network connectivity between two devices by transmitting Internet Control Message Protocol (ICMP) Echo Request messages and waiting for Echo Replies.
- **Execution:** C:\> ping 192.168.1.2



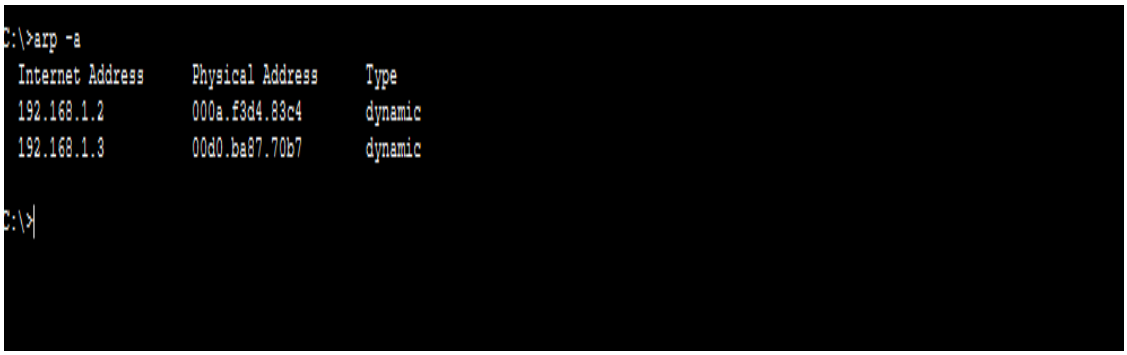
COMMAND 3: TRACERT

- **Purpose:** Determines the path that packets take to reach a destination by displaying each router hop along the route using ICMP Time Exceeded messages.
- **Execution:** C:\> tracert 192.168.1.2



COMMAND 4: ARP -A

- **Purpose:** Displays the current Address Resolution Protocol (ARP) cache table, which maintains mappings of Layer 3 IP addresses to Layer 2 physical MAC addresses.
- **Execution:** C:\> arp -a



COMMAND 5: NETSTAT

- **Purpose:** Displays active TCP connections, listening network ports, Ethernet statistics, and routing table information.

- **Execution:** C:\> netstat



4. COMMAND SUMMARY REFERENCE TABLE

CLI Command	Protocol / Layer	Primary Purpose	Key Diagnostic Value
ipconfig	TCP/IP (OS Layer)	Display local IP parameters	Confirm static/DHCP IP assignment, subnet mask, and default gateway.
ping	ICMP (Layer 3)	Test end-to-end connectivity	Measure latency, round-trip time, and packet loss percentage.
tracert	ICMP / IP (Layer 3)	Trace route path across hops	Locate intermediate routing bottlenecks or path failures.
arp -a	ARP (Layer 2.5)	Inspect IP-to-MAC cache	Verify dynamic address resolution and detect ARP spoofing.
netstat	TCP / UDP (Layer 4)	Display active connections	Monitor listening ports, active sockets, and transport session health.

5. CONCLUSION

This lab verified TCP/IP layer operations using CLI diagnostic utilities on a switched LAN: ipconfig confirmed static IP settings, ping verified zero packet loss, tracert showed single-hop local switching, arp -a resolved dynamic MAC addresses, and netstat monitored active sockets.